ELLK-Net: An Efficient Lightweight Large Kernel Network for SAR Ship Detection

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***Abstract*— ELLK-Net, an efficient, lightweight network with a large kernel, is proposed for synthetic aperture radar (SAR) ship detection. It addresses background variations, different ship scales, and noise interference challenges. ELLK-Net uses an anchor-free detector framework and sequentially decomposes large kernel convolutions to capture comprehensive global information and long-range dependencies. It adaptively selects convolution kernels on the basis of target characteristics, enhancing multiscale feature expression. A novel large kernel multiscale attention (LKMA) module is introduced to enhance interlayer feature fusion and semantic alignment, mitigating the impacts of overlapping ships and scattering noise. Structural reparameterization techniques optimize inference speed across devices without compromising accuracy. The experimental results on the SAR ship detection dataset (SSDD) and high-resolution SAR image dataset (HRSID) datasets demonstrate that ELLK-Net achieves impressive AP_{50} values of 95.6% and 90.6% for horizontal box detection and 89.7% and 79.7% for rotating box detection, respectively. The reparameterized detector exhibits a significant 48.7% FPS improvement on the Nvidia Jetson NX platform, indicating its suitability for edge computing deployment. The code is available at https://github.com/CHD-IPAC/ELLK-Net.**

***Index Terms*— Anchor-free, large kernel network, lightweight convolutional neural networks (CNNs), ship detection, synthetic aperture radar (SAR).**

I. Introduction

SHIP detection is vital in monitoring marine environments, providing essential technical support for maritime transportation management and decision-making [1], [2], [3], [4], [5], [6]. Synthetic aperture radar (SAR) is an advanced microwave sensor capable of high-resolution 2-D imaging [7]. Compared with optical, infrared, and other remote sensing technologies, SAR can operate day and night under all weather conditions, making it indispensable for maritime vessel monitoring.

In recent years, with the development of deep neural networks, convolutional neural network (CNN)-based object detectors have gradually replaced feature extraction methods based on manual features and have become the mainstream approach in computational vision. CNN-based object detectors can be divided into two main types: anchor-based methods and anchor-free methods. The anchor-based approach involves predefining anchors with different scales and aspect ratios for each pixel. Feature maps are generated via a feature extraction network, and a detection head selects the anchor containing the target. The method calculates the category confidence, adjusts the position and category of the anchor, and selects positive and negative samples on the basis of the intersection-over-union (IoU) metric. Redundant anchors are filtered out via nonmaximum suppression (NMS) to complete the detection task. Prominent detectors in this category include the region-based CNN (RCNN), Faster R-CNN [8], single-shot multibox detector (SSD), you only look once (YOLO) series, and RetinaNet [9].

In the field of SAR ship detection, Sun et al. [10] proposed a YOLO-based SAR ship detector based on bidirectional feature fusion and angle classification. Lin et al. [11] introduced SER Faster R-CNN, which enhances detection performance through the squeeze-and-excitation (SE) mechanism. Bai et al. [12] designed a novel SAR ship detection network that achieves superior accuracy through feature enhancement and reconstruction. Although anchor-based detectors significantly enhance SAR ship detection performance, they rely on prior information such as scale, features, and categories when designing anchor boxes.

The anchor-based detectors demonstrate high accuracy but have limited generalizability across diverse datasets because of the fixed size of the anchor boxes. Simultaneously, anchor box settings incur substantial computational costs and exacerbate positive-negative sample imbalance issues. In response to these limitations, anchor-free object detection methods have been developed. These methods can be categorized into keypoint combination, centerpoint regression, and transformer methods.

CornerNet adjusts prediction box positions via corner pairs, accomplishing target detection without anchor boxes. Ge et al. [13] proposed YOLOX, a detector that limits positive samples to a specific grid region of the target center. They also decoupled the regression and classification branches in detection heads, which accelerated model convergence and improved accuracy. Tian et al. [14] used the fully convolutional one-stage object detector (FCOS), a pixel-by-pixel prediction detector

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based on a convolutional network. This method considers center points within the true box range as positive samples. Zhou et al. [15] proposed CenterNet, an object detection method that avoids the need to partition input images. For the first time, DETR [16] successfully integrated transformers for object detection. Deformable DETR [17] uses multiscale deformable attention to replace DETR's attention module, improving target detection. However, transformer-based detection methods greatly increase computational and spatial complexity, and the detection accuracy for small objects needs further improvement.

For SAR ship detection, Gao et al. [18] proposed an anchor-free detector with intensive attention feature aggregation to improve multiscale detection performance. Bai et al. [19] designed a global context-guided feature balance pyramid structure to enhance the network's multiscale feature representation capability by learning global context information. Fu et al. [20] proposed an attention-oriented balance pyramid based on FCOS to improve small-ship performance.

SAR imaging quality, ship characteristics, and environmental conditions affect the accuracy of ship detection methods. The diverse shapes and structures of ships cause uneven scattering center distributions and visual disparities, leading to potential false detections in complex inshore regions. Performance varies greatly for different ship sizes at various resolutions, requiring robust multiscale feature expression capabilities in the detection network. Most SAR ship detectors are designed as general target detectors with a backbone network, a neck using a feature pyramid network (FPN), and a head. The FPN combines multiscale features but cannot enrich feature diversity after fusion, limiting performance improvement, especially for anchor-free SAR ship detection. Enhancing the multiscale expression of fine-grained features through different convolution sizes is crucial.

Many deep network-based SAR target detectors use CNNs, but they have limited receptive fields and struggle to model long-range pixel dependencies. This results in insufficient expression ability for fine-grained multiscale features. Existing backbone designs achieve large receptive fields through numerous small convolution kernels, but this requires excessive parameters and computation. Deeper network layers provide only marginal gains in the effective receptive field. A visual transformer (ViT) uses self-attention to model long-range dependencies and obtain a global receptive field. However, ViT-based methods in SAR ship detection present issues such as high training costs, slow convergence, and poor small target detection.

Recently, there has been a successful revival of modern large kernel convolutional networks such as ConvNeXt [21] and RepLKNet [22], which have validated that simply increasing the convolution kernel size can improve network performance. However, in SAR ship detection, directly using large kernel convolution leads to false and missed detection of smaller targets because of an excessive focus on the background and larger targets. Moreover, large kernel convolution can hinder training and deployment, resulting in hardware insufficiency and reduced performance. Therefore, designing an efficient SAR ship detector based on large kernel convolution for fine-grained multiscale target characterization while also optimizing inference execution speed across diverse devices is a valuable research problem.

To address the aforementioned problems, this article proposes an efficient lightweight large kernel network, ELLK-Net, for SAR ship detection. The main contributions of this article are summarized as follows.

1) We present an adaptive large kernel selection (ALKS) module to enhance the effective receptive field of the detector backbone network and improve fine-grained multiscale feature representation. ALKS is stacked at multiple levels to construct an ALKS backbone network (ALKS-Net). Large kernel convolutions capture global information and model long-range dependencies between the background and target. By adaptively selecting large kernels in various receptive fields, ALKS-Net enhances feature expression for different-scale objects.
2) To address overlapping ship targets and noise interference, we propose a large kernel multiscale attention feature enhancement pyramid (LKA-FEP) module in the detector structure. This module enhances interlayer feature fusion and semantic alignment across multiple scales.
3) Building on the FCOS framework, we incorporated structure reparameterization to decouple the training and inference stages of large kernel convolution. This created an efficient, lightweight network, ELLK-Net, customized for SAR ship detection. ELLK-Net reduces the number of parameters while maintaining accuracy and increasing the inference speed across devices.

Our network was validated on the SAR ship detection dataset (SSDD) and high-resolution SAR image dataset (HRSID) and demonstrated exceptional precision and speed in both horizontal and rotational box detection tasks. This approach presents a viable solution for deployment on edge devices. This article is organized as follows. Section II presents the details of the proposed SAR ship detector. Section III describes the benchmark dataset and experimental setup. Section IV discusses the specifics and results of the experiment. Finally, Section V concludes the article.

II. Proposed Method for SAR Ship Detection

SAR images exhibit significant differences between inshore scenes and offshore scenes. Ships are easily impacted by scattering noise. Significant variation exists in ship sizes and resolutions, requiring high multiscale feature expression and fusion. Anchor-free detectors directly predict keypoints of bounding boxes, relying heavily on fine-grained feature expressions from backbones. Existing SAR ship detectors prioritize either accuracy or speed, making it difficult to balance both. ELLK-Net uses sequentially decomposed large kernel convolution for long-range dependence modeling, with adaptive selection of convolution kernels for improved multiscale feature expression. The network also uses a large kernel multiscale attention (LKMA) module and an improved feature pyramid for enhanced interlayer multiscale feature fusion and semantic information alignment, reducing the impacts of overlapping ships and scattering noise on detection accuracy.

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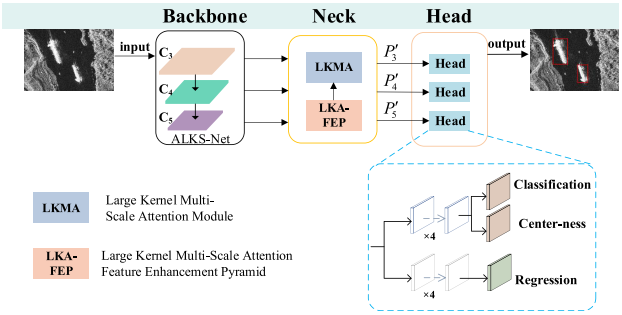


Fig. 1. Overall architecture of ELLK-Net consists of three parts. 1) ALKS-Net as the backbone. 2) LKA-FEP and LKMA as the neck. 3) Detection head.

Structural reparameterization decouples large kernel convolution training and inference stages, improving inference speed across devices while maintaining accuracy.

A. Overall Architecture

Fig. 1 shows the overall architecture of ELLK-Net, which is organized into three primary components: a backbone network, a neck, and a head. The backbone network uses ALKS-Net, which outputs multilevel feature maps from the last three stages of ALKS-Net: C_3, C_4, and C_5. Then, the multilevel feature maps are input to LKA-FEP for feature fusion, and P_3'–P_7' are the feature map outputs. Finally, the detection head uses the feature maps of P_3'–P_7' as the prediction layer to train and predict the feature maps output by different feature layers and obtain the final detection results. In ELLK-Net, we use five pyramid feature layers (P_3'–P_7'), where P_6' and P_7' are generated by applying a 3×3 convolutional layer with a stride of 2 on P_5' and P_6', respectively. With respect to the FCOS framework, ELLK-Net's detection head has three branches: classification, regression, and centrality. Classification and centrality are two different small branches on the same branch. Each branch first goes through a combination of four Conv2d + GN + ReLU modules, followed by a convolutional layer with a 3×3 convolution kernel size and a stride of 1 to obtain the final prediction result. If rotating object detection is needed, only the angle regression branch is added to the detection head.

B. Adaptive Large Kernel Selection Network

SAR ship detection faces several challenges, such as target overlap, noise sensitivity, and varying feature distributions. The anchor-free method, which relies heavily on the semantic information expression of the feature map, cannot enrich feature diversity through interlayer fusion. The commonly used CNN backbone has a limited receptive field and rarely models long-range dependencies, hindering further performance improvement. Inspired by [21], [22] and [23], we build an ALKS network (ALKS-Net) via an ALKS module. The structure of ALKS-Net is shown in Fig. 2. ALKS-Net adopts a hierarchical design with four stages to facilitate subsequent feature pyramid input. Each stage reduces the input feature map resolution and expands the receptive field layer by layer. Within each stage, the ALKS blocks are stacked to deepen the network and enhance the representation ability. Each

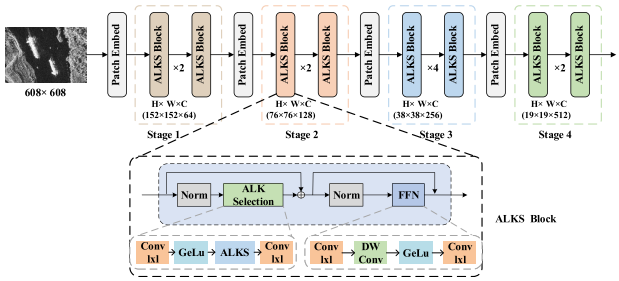


Fig. 2. Structure of the ALKS-Net.

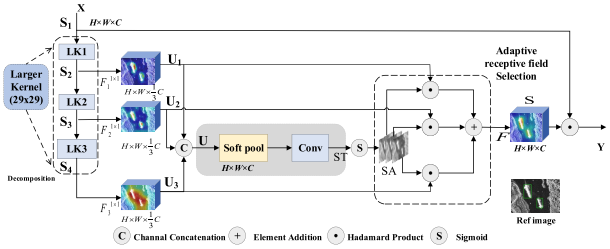


Fig. 3. Structure of the ALKS module.

ALKS block contains an ALK selection subblock for adaptive receptive field adjustment and a feedforward network (FFN) subblock for channel blending and feature refinement.

The ALKS module is the core of ALKS-Net and is used in the ALK selection subblock, as shown in Fig. 3. The advantage of large kernel convolutions is that they can obtain a transformer-like global receptive field and model long-range dependencies between pixels. Simply using large kernel convolution will lead to increased computational cost. In this work, a 29×29 large convolution kernel is decomposed into several depthwise separable convolution (DW) kernels with different receptive fields via sequential decomposition. Different kernels are adaptively used through the spatial kernel selection mechanism to adjust the receptive field of each object in the spatial dimension. Unlike SKNet, which adaptively adjusts the receptive field size in the channel dimension, the ALKS module is spatially adjusted, making it more suitable for SAR ship detection. Considering three ship scales in the SAR image—small, medium, and large—we decompose large kernel convolutions into 3×3 normal convolutions (LK_1), 5×5 depthwise dilated convolutions (LK_2), and 7×7 convolutions built by a dilated multibranch module (DMM) (LK_3). First, the input feature map X is fed into three sequential decomposition convolutions to obtain spatial feature maps S_i of different receptive fields

$$S_1 = X, \quad S_{i+1} = \mathrm{LN}(\sigma_{\mathrm{ge}}(\mathrm{LK}_i(S_i))), \ i \in [1, 3] \tag{1}$$

where σ_{ge} represents the Gaussian activation function, LN represents layer normalization, and LK_i represents the three sequentially decomposed convolutions with different receptive fields. The decomposed spatial feature maps are subsequently fused via a 1×1 convolution ($F_i^{1\times1}$)

$$U_i = F_i^{1\times1}(S_{i+1}), \quad i \in [1, 3]. \tag{2}$$

Considering the multiscale phenomenon of SAR ship targets, we employ a spatial selection mechanism to dynamically select

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the appropriate receptive field from LK_1, LK_2, and LK_3 with different scales to increase the ability of the network to express fine-grained multiscale features. The implementation of this adaptive selection mechanism is as follows. First, the spatial features from different receptive fields in LK_1, LK_2, and LK_3 are concatenated

$$U = \text{concat}(U_1; \ldots; U_i), \quad i = 3. \tag{3}$$

On the channels, important spatial information in the feature map is preserved by SoftPool, and these spatially pooled features are connected. The network passes through a 1×1 convolutional layer to obtain the spatial feature ST with the number of channels aligned

$$\text{ST} = \text{conv}(\text{Softpool}(U)). \tag{4}$$

Afterward, the sigmoid activation function (denoted by σ_{sig}) is applied to spatial feature ST to obtain independent spatial selection weights corresponding to each sequentially decomposed large convolution kernel

$$\text{SA}_i = \sigma_{\text{sig}}(\text{ST}). \tag{5}$$

Then, the feature maps from different receptive fields of the LK_1, LK_2, and LK_3 convolutions are weighted by their corresponding spatial selection weights and fused via the convolution layer F to obtain the overall large kernel convolution spatial attention weight K, where $\odot$ represents the Hadamard product and N is the number of channels

$$K = F\left(\sum_{i=1}^{N}(\text{SA}_i \odot U_i)\right). \tag{6}$$

Finally, the output of the ALKS module can be obtained by the Hadamard product of the input feature map X and the overall large kernel convolutional spatial attention weight K

$$Y = X \odot K. \tag{7}$$

C. LKMA Feature Enhancement Pyramid

SAR images face challenges in aligning and fusing semantic information because of scattering noise and overlapping targets with varying feature distributions. Enhancing interlayer multiscale feature fusion and semantic alignment within the neck partial feature fusion network is necessary to improve detection accuracy.

In SAR ship detection, the FPN is used for bottom-up feature fusion. However, scattering noise in SAR images can cause misaligned target features and spatial information mismatches, leading to inaccurate positioning and hindering improvements in detector accuracy. To address this, we propose a LKA-FEP, inspired by [24] and [25]. We mitigate feature-spatial mismatch by incorporating a feature alignment module (FAM) during upsampling. Additionally, we employ the LKMA module to capture the global context and establish long-range dependencies among multiscale features via a grouped multibranch structure. This reduces the impact of scattering noise on both multiscale feature fusion and semantic alignment between network layers.

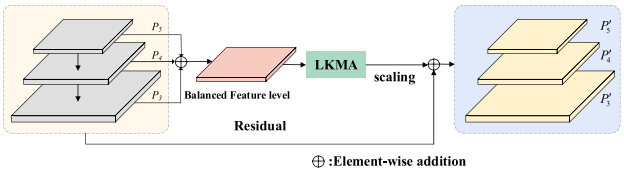


Fig. 4. Structure of the LKA-FEP.

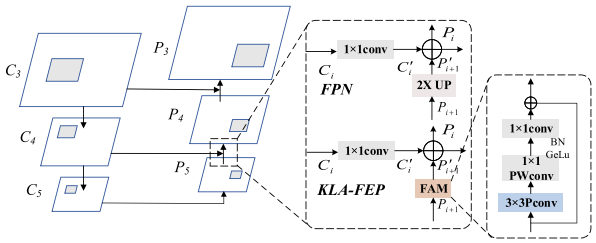


Fig. 5. Structural comparison of FPN and LKA-FEP.

As shown in Fig. 4, LKA-FEP balances the semantic information of features at different levels by aggregating and averaging, unlike the general FPN. The outputs (C_3, C_4, C_5) of ALKS-Net are obtained via feature fusion through a top-down path (P_3, P_4, P_5), which is defined as P_L. Then, the multilayer feature maps are adjusted and aggregated by the balanced feature level (BFL), bilinear interpolation is used to adjust the size of P_5, which is smaller than P_4, and max pooling is used to adjust the size of P_3, which is larger than P_4. The P_3 and P_5 space dimensions are adjusted to the same size as P_4. LKA-FEP uses large kernel convolution to obtain long-range dependencies of feature maps and obtains multiscale features across dimensions through channel grouping and multibranch structures. This process can be described as follows:

$$P'_L = \text{Scaling}\left(\text{LKMA}\left(\frac{1}{L}\sum_{L_{\min}}^{L_{\max}} P_L\right)\right) + P_L. \tag{8}$$

L denotes the number of feature maps, and LKMA can be formulated as follows:

$$Y^{\text{LKMA}}(X) = \sigma_{\text{sig}}(\text{SoftPool}(X_{1\times1}(LK(X))) + \text{Softpool}(X_{3\times3}(LK(X)))) + X \tag{9}$$

where LK represents the 23×23 large convolution kernels, $X_{1\times1}$ and $X_{3\times3}$ represent the 1×1 branch and 3×3 branch, respectively, SoftPool represents 2-D global soft pooling, and σ_{sig} represents the sigmoid function.

In LKA-FEP, we design FAMs for upsampling, as shown in Fig. 5. The FAM is composed of 3×3 PConv, 1×1 PWConv, and 1×1 Conv, which can align the feature maps of the current layer with the features of the previous layer. This approach is conducive to fusing cross-scale information and improving the ability of the network to detect objects of different scales.

The convolutional block attention module (CBAM), coordinated attention, and efficient channel attention (ECA) have shown the benefits of grouping channel dimensions into multiple subfeatures. This approach results in an even distribution of semantic features in each feature group, and cross-dimensional

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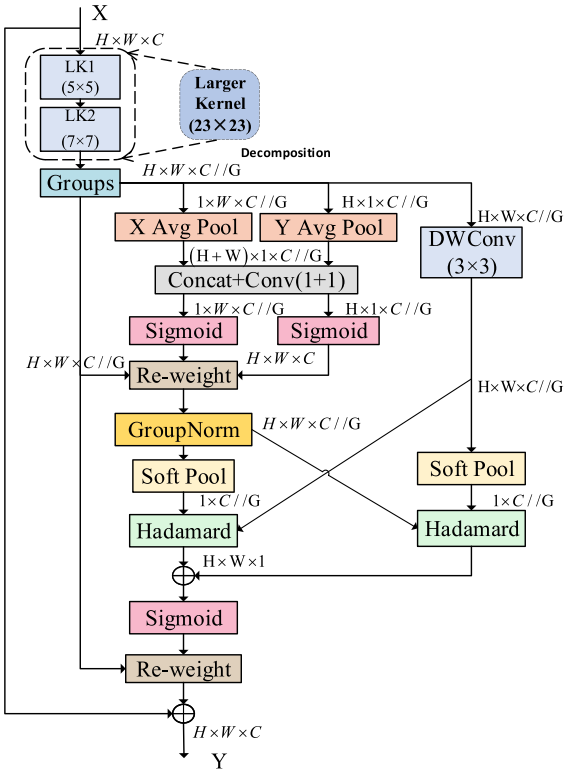


Fig. 6. Structure of the LKMA module.

interactions contribute to channel or spatial attention prediction. The LKMA module structure inspired by [24] is shown in Fig. 6. The LKMA module works by first decomposing the aggregated features of the BFL output through a 23 × 23 large kernel convolution. This results in a feature map with a large receptive field, which is then divided into G groups according to the channel dimension. In each group, the LKMA constructs parallel subnetworks with different receptive fields via 1 × 1 and 3 × 3 convolution branches in the horizontal and vertical directions, respectively. This allows for the capture and aggregation of multiscale semantic information. Cross-channel information is modeled by two 1-D global average pooling operations in the horizontal and vertical channel directions, together with branches using a 3 × 3 DW convolution to capture multiscale feature representations. In this way, location information can be embedded into channel attention, thus reducing computational costs.

Unlike the attention weights learned by the CBAM, ECA, and other methods through simple averaging, the LKMA uses cross-space learning to fuse and merge attention maps from subnetworks. Spatial information from different dimensions in the 1 × 1 branch and 3 × 3 branch networks is aggregated by the Hadamard product to generate two spatial feature attention maps. Instead of using MaxPool as in [24], we employ SoftPool to retain more information in downsampling activation maps, thus aiding feature alignment.

D. Structural Reparameterization of Large Kernel Convolutions

Most existing SAR ship detectors emphasize model training performance while ignoring the significance of inference

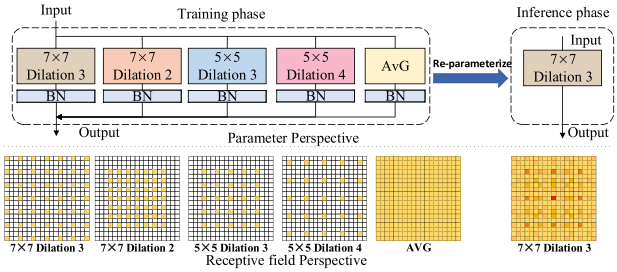


Fig. 7. Structure of the DMM.

performance. Structural reparameterization uses mathematical transformations to convert complex neural networks into simpler networks, accelerating inference while preserving accuracy. Through structure reparameterization, training, and inference model structures can be decoupled [22], [26], improving the suitability of graphics processing units (GPUs) and edge devices with dense computational architectures. Compared with quantization, pruning, distillation, and other compression methods, structure reparameterization ensures accuracy and speed inference, making it suitable for large kernel network training and deployment.

During ELLK-Net model training, the 7 × 7 large kernel convolution with sequential decomposition is replaced by the DMM. The DMM, depicted in the upper section of Fig. 7, employs a convolution and average pooling multibranch topology with varying dilation rates and convolution kernels. The lower part of Fig. 7 illustrates receptive field changes for different convolutions. During weight fusion, ignoring input pixels is equivalent to creating additional holes in the Conv kernel. This allows the small-kernel dilated convolution to be converted to a large kernel normal convolution (dilation rate $r = 1$). Let k be the kernel size of the unfolded layer, and by inserting voids, the kernel size of the corresponding nonunfolded layer is $(k-1)r+1$, referred to as the equivalent kernel size. However, different dilated convolutions with the same equivalent kernel size can also be converted equivalently. An average pooling layer with kernel size K, stride S, and input channel C can be converted into a convolution $F' \in R^{C\times C\times K\times K}$ with the same kernel size and stride

$$F'_{d,c,:,:} = \begin{cases} \dfrac{1}{K^2}, & \text{if} \quad d = c \\ 0, & \text{elsewise.} \end{cases} \tag{10}$$

During model inference, the DMM can be reduced to a sequential decomposition of 7 × 7 large kernel convolutions through weight fusion, incurring no additional inference time cost.

E. Loss Function

ELLK-Net uses a multitask loss function for optimization, which is detailed below

$$L_{\text{loss}} = \frac{1}{N_{\text{pos}}}\sum_{i=1}^{n} L_{\text{cls}}\left(p_i, p_i^*\right) + \frac{\lambda}{N_{\text{pos}}}\sum_{i=1}^{n} L_{\text{reg}}\left(\vartheta_i, \vartheta_i^*\right) \tag{11}$$

where L_{cls} and L_{reg} are classified losses and regression losses, respectively. N_{pos} is the number of samples, and p_i^* and ϑ_i^* are

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the ground-truth labels of the sample and the sample bounding box, respectively. p_i denotes the probability that the sample belongs to the foreground. ϑ_i is the prediction value of the bounding box

$$\mathrm{FL}(p_t) = -\alpha_t(1-p_t)^{\gamma}\log(p_t) \tag{12}$$

where

$$p_t = \begin{cases} p_t, & \text{if } y = 1 \\ 1-p_t, & \text{otherwise.} \end{cases} \tag{13}$$

p_t is the probability value predicted by the model, a_t is the weight coefficient balancing positive and negative samples, and γ is an adjustable parameter to solve the imbalance of difficult and easy-to-distinguish samples.

The IoU loss is used to compute the regression loss and can be expressed as follows:

$$\mathrm{IoU}_{\mathrm{loss}} = 1 - \frac{A_{\mathrm{pred}} \cap A_{\mathrm{gt}}}{A_{\mathrm{pred}} \cup A_{\mathrm{gt}}}. \tag{14}$$

A_{gt} represents the true bounding box, and A_{pred} represents the predicted bounding box.

III. Experimental Details

A. Datasets and Settings

The experiments in this article utilize two datasets: SSDD [27] and HRSID [28]. The SSDD dataset consists of images with resolutions ranging from 1 to 15 m, widths ranging from 214 to 668 pixels, and heights ranging from 160 to 526 pixels. It contains a total of 1160 images with 2456 ships, of which 928 images are used for training and 232 images are used for testing. To ensure comparability, all the input images in this article are uniformly resized to 608 × 608 pixels. The HRSID dataset is derived by cropping sections from 136 panoramic high-resolution SAR images, resulting in 5604 SAR images sized 800 × 800 pixels each. This dataset comprises 3642 training images and 1962 testing images. Both datasets include rotation annotations that facilitate the experiments.

The effectiveness of ELLK-Net is verified through experiments conducted on an RTX 4090 GPU with the Ubuntu 22.04 operating system. The code is implemented via PyTorch. We utilize MMdetection [29] to standardize the experimental benchmarks and ensure a fair comparison. Our proposed method employs stochastic gradient descent (SGD) for network optimization and adopts a cosine linear learning strategy, utilizing a batch size of eight images and training for 100 epochs. The initial learning rate is set to 0.0025, whereas the weight decay and momentum are set to 0.0001 and 0.9, respectively.

B. Evaluation Metrics

This article uses average precision (AP), AP_{50}, AP_{75}, AP_s, AP_m, and AP_l as evaluation metrics, which have the same meanings as in the COCO dataset, to evaluate the proposed method comprehensively. AP is the average precision as the IoU = 0.5:0.05:0.95; AP_{50} is the average accuracy at an IoU threshold of 0.5; and AP_{75} is the average accuracy at an IoU threshold of 0.75, a more stringent evaluation metric reflecting the object localization accuracy. AP_s, AP_m, and AP_l evaluate the detection accuracy for objects of different sizes, with AP_s representing small targets (area $< 32^2$ pixels), AP_m representing medium targets (32^2 < area < 64^2 pixels), and AP_l representing large targets (area $> 64^2$ pixels). The precision and recall are defined as

$$\begin{aligned} \mathrm{Precision} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \\ \mathrm{Recall} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \\ \mathrm{AP} &= \int_0^1 P(R)dR \end{aligned} \tag{15}$$

where TP represents the number of correctly detected samples, FP represents false positives, and FN represents missing targets. AP is the main evaluation index for evaluating the comprehensive performance of the model, and $P(R)$ is the precision–recall curve.

The number of floating point operations per second (FLOPs), the number of parameters (Params), and the number of frames per second (FPS) are used as evaluation metrics for detector complexity.

C. Comparison Experiment

In this section, two datasets, SSDD and HRSID, are used to compare the performances of ELLK-Net and several state-of-the-art methods. These methods include YOLOv3 [30], YOLOv5, YOLOX [13], Faster R-CNN [8], FCOS [14], RetinaNet [9], FoveaBox [31], Libra R-CNN [25], DETR [16], HRSDNet [28], FEPS-Net [12], FBR-Net [20], LSKNet [23], KEA-Net [32], and MSD-Net [33].

Comparisons of different detection methods with respect to the SSDD dataset are listed in Table I, and the corresponding precision–recall (PR) curves are shown in Fig. 8. Table I shows that ELLK-Net significantly reduces the number of parameters and computations while achieving the best detection results. Compared with those of the Libra R-CNN, the parameters are reduced by 59.6%, whereas the AP_{75} increases by 3.9%. However, the AP_m of ELLK-Net is 1.2% lower than that of the Libra R-CNN. This is because the Libra R-CNN scales and aggregates features from different layers to the middle layer size in the feature pyramid (P_4'), enhancing the performance in terms of the AP_m. However, our method increases AP_s and AP_l by 1.8% and 3.4%, respectively, which indicates a more balanced effect on object detection across scales. The performance of our method surpasses Faster RCNN by 1.7% in terms of AP_s detection. This is because, as an anchor-based detector, Faster R-CNN requires many hyperparameters to be set. Additionally, the small size of ship targets in SAR images causes anchor box mismatches, which affects detection accuracy. Compared with our method in terms of AP_s, AP_m, and AP_l, FoveaBox, an anchor-free detector, yields differences of 6.6%, 1%, and 1.5%, respectively. This disparity arises from the direct regression of the anchor-free detector on the feature map, which relies heavily on the semantic information expression ability of the map. Although most of these

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TABLE I
COMPARISON OF DIFFERENT DETECTION METHODS ON THE SSDD DATASET

method	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l	FLOPs (G)	Params (M)
FoveaBox [31]	0.587	0.911	0.672	0.506	0.712	0.709	82.2	37.1
Faster R-CNN [8]	0.621	0.934	0.721	0.555	0.722	0.723	83.6	41.2
Libra R-CNN [25]	0.623	0.930	0.707	0.554	**0.734**	0.690	83.9	41.4
RetinaNet [9]	0.602	0.942	0.682	0.539	0.698	0.707	73.8	36.1
FCOS [14]	0.601	0.936	0.689	0.527	0.711	0.676	72.4	31.8
YOLOv3 [30]	0.479	0.927	0.410	0.427	0.570	0.569	**5.93**	**3.7**
YOLOX [13]	0.531	0.922	0.547	0.512	0.595	0.504	12.2	8.9
YOLOv5n	0.595	**0.972**	-	-	-	-	25.2	7.4
DETR [16]	0.557	0.922	0.616	0.488	0.682	0.682	71.5	40.0
FBR-Net*[20]	-	0.941	0.591	-	-	-	-	32.5
KEA-Net*[32]	0.622	0.963	0.7113	0.556	0.706	0.720	-	-
LSKNet [23]	0.570	0.936	0.640	0.504	0.678	0.701	60.6	21.8
HRSDNet*[28]	0.611	0.939	0.701	0.566	0.677	0.588	-	37.2
FEPS-Net [12]	0.599	0.960	0.675	0.551	0.682	0.706	71.1	37.3
Our method	**0.639**	0.956	**0.746**	**0.572**	0.722	**0.741**	25.4	16.9

Note: Since the source codes of some references are unavailable, the detection results of methods denoted * are from these references

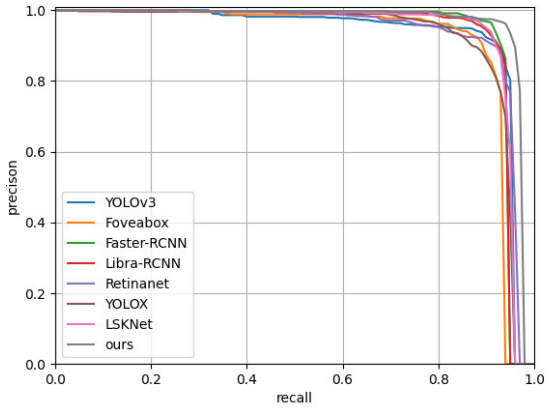


Fig. 8. PR curves of different detection methods for SSDD.

anchor-free detectors use CNNs for feature extraction, their limited receptive field makes it difficult to model long-range dependencies between pixels and express fine-grained multiscale features. However, our method improves the fine-grained multiscale feature representation ability of the feature map by adaptively selecting large kernel networks. DETR based on transformers that performs well in terms of AP_m and AP_l but has little effect in AP_s. This is because transformers obtain a global receptive field through self-attention, which focuses on medium and large objects while ignoring small objects, resulting in low detection performance for small objects. Compared with the YOLOv3 lightweight model, although the number of parameters of ELLK-Net increases by 13.23 M, the AP increases by 16%, which indicates that our method can achieve efficient detection accuracy with fewer parameters.

The visualized feature maps in Figs. 11 and 12. A visual analysis of the SSDD detection results is shown in Figs. 13–16. Missing and false detections are represented by yellow and blue circles, respectively. Compared with the other methods, ELLK-Net generates fewer missed detections in various scenes, such as small ships, densely arranged ships, and inshore interference. Faster R-CNN, Libra R-CNN, and FoveaBox perform well in inshore interference and densely arranged ship scenes but have many missed detections in small-ship scenes. YOLOv3 and YOLOX can detect small ships well, but false and missed detections occur for large, dense targets. RetinaNet exhibits many false detections in scenes with small ships and missed detections in scenes with dense ship arrangements and inshore interference. Compared with the FoveaBox and YOLOX anchor-free detectors, our method performs better across the scenes. This is because our method uses large kernel convolution to model long-range dependency between background and objects via ALKS-Net, enhancing the fine-grained multiscale feature expression of the network. Additionally, LKA-FEP is used for consistent feature fusion across different layers, ensuring consistent detection accuracy for multiscale targets in diverse scenes.

Table II shows the results of comparative experiments on the HRSID dataset. Compared with HRSDNet, our method reduces the number of parameters by 54.6%. Although HRSDNet performs well for small object detection, it underperforms for large objects. In contrast, our method improves the AP by 5.6% on AP_l. This improvement is due to LKA-FEP, which balances multilevel semantic information and learns context to enhance multiscale ship detection, yielding more balanced results across different scales. Notably, the input of HRSDNet is 1000×1000, which can improve the final detection accuracy. Compared with the FoveaBox, YOLOX, and FCOS anchor-free detectors, our method achieves significant accuracy improvements. This is due to the improved fine-grained multiscale feature expression of feature maps via the ALKS used in our method. Our method also increases the AP by 23.3% compared with the YOLOv3 lightweight model. The corresponding PR curves are shown in Fig. 9, and the detection results for the HRSID dataset are shown in Figs. 17 and 18.

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TABLE II
COMPARISON OF DIFFERENT DETECTION METHODS ON THE HRSID DATASET

method	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l	FLOPs (G)	Params (M)
FoveaBox [31]	0.633	0.862	0.714	0.644	0.632	**0.391**	232.8	37.1
Faster R-CNN [8]	0.659	0.874	0.766	0.668	**0.674**	0.247	211.3	41.2
Libra R-CNN [25]	0.659	0.866	0.765	0.667	0.670	0.223	212.3	41.4
RetinaNet [9]	0.649	0.884	0.734	0.662	0.658	0.288	209.1	36.1
FCOS [14]	0.647	0.890	0.736	0.662	0.659	0.258	198.5	31.8
YOLOv3 [30]	0.435	0.775	0.442	0.443	0.466	0.058	**16.8**	**3.7**
YOLOX [13]	0.619	0.881	0.707	0.637	0.586	0.071	34.1	8.9
YOLOv5n	0.645	0.886	-	-	-	-	70.3	7.4
DETR [16]	0.458	0.719	0.516	0.457	0.525	0.101	199.8	40.0
MSD-Net* [33]	0.606	0.855	0.679	0.614	0.655	0.320	118.4	38.2
LSKNet [23]	0.630	0.883	0.713	0.646	0.619	0.150	171.6	21.8
HRSDNet* [28]	0.686	0.884	0.790	0.696	0.700	0.252	-	37.2
FEPS-Net [12]	0.652	0.885	0.733	0.662	0.676	0.337	194.2	36.5
Our method	**0.668**	**0.906**	**0.760**	**0.689**	0.660	0.308	67.2	16.9

Note: Since the source codes of some references are unavailable, the detection results of methods denoted * are from these references

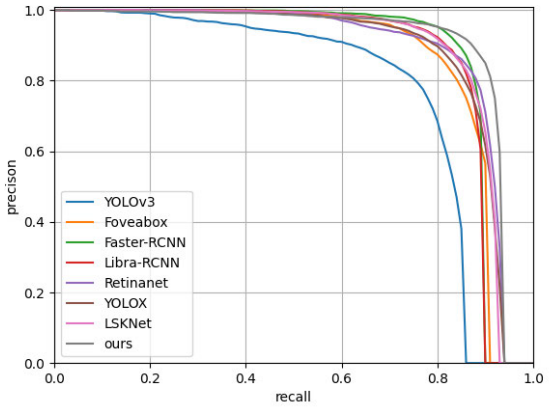


Fig. 9. PR curves of different detection methods for HRSID.

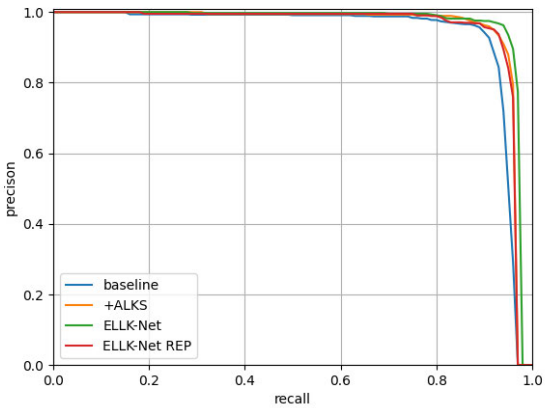


Fig. 10. Improved PR curves of different modules.

In Figs. 16 and 17, land occupies most of the area in the inshore interference scene and the inland complex interference scene. These scenes contain considerable interference noise, which has similar characteristics to ships, making it difficult to detect ships. The YOLOX anchor-free detector produces many false detections due to land noise interference, mistakenly identifying noise as ships. Other methods, such as Faster R-CNN, Libra R-CNN, FoveaBox, and RetinaNet, also have many missed detections and few false detections. In contrast, ELLK-Net outperforms the comparison methods in both scenarios. The reason behind this is that our method incorporates ALKS-Net to effectively capture long-range dependencies between the background and object, enhancing the precise multiscale feature representation of the network. It also uses LKA-FEP for feature fusion between layers, maintaining good detection accuracy for multiscale targets across scenes. In scenes with numerous small ships, as shown in Fig. 18, the existing methods produce varying degrees of missed detections, and YOLOv3 and YOLOX yield a small number of false detections. However, ELLK-Net performs well in small-ship detection, with only a few missed detections.

D. Ablation Studies and Analysis

To verify the effectiveness of the proposed modules, we conduct a series of ablation experiments on the SSDD dataset. In this section, we evaluate and analyze the proposed method using the FCOS as the baseline network. Detection performance is analyzed via various metrics. To ensure fairness, we set the parameters to be the same across all methods for a quantitative comparison.

Table III shows that ALKS-Net and LKA-FEP contribute differently to the detection accuracy. Compared with the baseline network, the proposed network achieves 4.2% and 6.5% gains for AP and AP_{75}, respectively. The proposed method performs 6.0% and 6.8% better than the baseline for AP_s and AP_l, respectively, indicating significantly improved accuracy for small and large ships and proving the proposed method's advantages in multiscale ship detection. Moreover, the network accuracy changes little before and after reparameterization, whereas the numbers of parameters and calculations decrease. The corresponding PR curve (Fig. 10) visualizes the gain of each module.

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TABLE III
RESULTS OF THE MODULE ABLATION STUDIES

ALKS	LKA-FEP(LKMA)	LKA-FEP(EMA)	REP	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l	Params (M)	FPS (img/s)
×	×	×	×	0.601	0.936	0.689	0.527	0.711	0.676	31.8	90.9
✓	×	×	×	0.622	0.943	0.736	0.544	0.703	0.708	17.1	-
✓	×	✓	×	0.631	0.958	0.734	0.566	0.714	0.673	17.2	-
✓	✓	×	×	**0.643**	**0.960**	**0.754**	**0.587**	**0.729**	**0.744**	17.2	99.3
✓	✓	×	✓	0.639	0.956	0.746	0.572	0.722	0.741	**16.9**	**110.6**

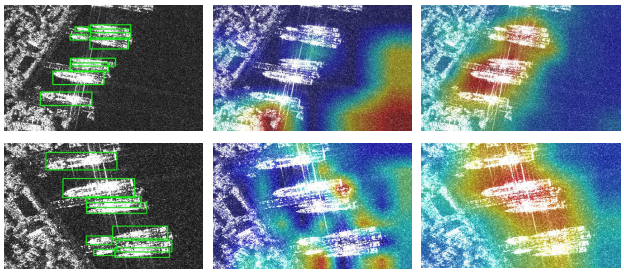


Fig. 11. Results of visualization of inshore interference scenes. (a) Ground truth. (b) FPN feature map visualization. (c) LKA-FEP feature map visualization.

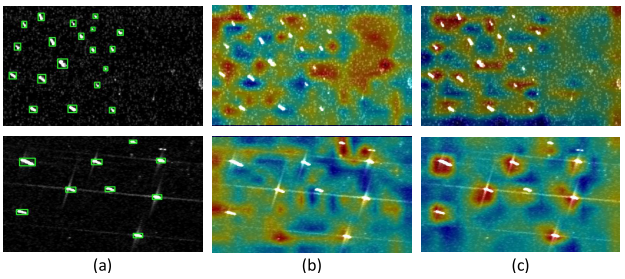


Fig. 12. Results of the visualization of the small-ship scene. (a) Ground truth. (b) FPN feature map visualization. (c) LKA-FEP feature map visualization.

After the feature extraction network is replaced with ALKS-Net, the number of parameters is reduced by 53.9%. Table III shows that AP_s and AP_l increased by 1.7% and 3.2%, respectively, whereas AP_m decreased by only 0.8%. Compared with ResNet50, the long-range dependence between the background and targets is modeled via large kernel convolution. Large convolutional kernels with different receptive fields are selected adaptively, thus benefiting the balance of detection effects across scales.

LKA-FEP replaces the FPN, and adopting the LKMA module increases the number of parameters by only 0.1 M and significantly improves all accuracy metrics. Table III shows that the gains of LKA-FEP for AP_s, AP_m, and AP_l are 4.3%, 2.6%, and 3.6%, respectively. Generally, pyramid-based detectors improve the AP_s and AP_m for small/medium objects but decrease the AP_l for large objects. However, LKA-FEP balances these metrics well by leveraging large kernel attention and feature alignment to balance multilevel semantics and learned context information, enhancing multiscale ship detection. This is especially critical for anchor-free detectors. The visualized feature maps in Figs. 11 and 12 revealed that the FPN does not fully focus on ship features under offshore strong scattering noise, whereas the LKA-FEP is fully focused and extracts more complete and clearer features than the FPN does. In offshore small-ship scenes, the attention of the FPN deviates greatly from the actual ship positions, whereas LKA-FEP locates small ships more accurately.

In Table III, the ablation experiments reveal that LKA-FEP does not use large kernel convolution and adopts the efficient multiscale attention (EMA) module. These experiments show that all accuracy indicators can be significantly improved by adding large kernel convolutions to the attention mechanism. The gains of LKA-FEP for AP_s, AP_m, and AP_l are 2.1%, 1.5%, and 7.1%, respectively. This is because SAR ship images contain more small- and medium-sized targets than medium- and large-sized targets do, and general attention focuses on small targets. A large kernel convolution can obtain a larger receptive field, allowing attention to be balanced for targets at different scales. REP indicates whether structure reparameterization is used. Table III shows that using reparameterization can further reduce the number of parameters while maintaining accuracy, and the FPS is improved by approximately 11.4%. Compared with that of the baseline network, the AP of ELLK-Net after reparameterization increases by 3.8%, the number of parameters decreases by 53.9%, and the FPS increases by 21.7%. This fully demonstrates the effectiveness of the proposed network.

E. Inference Performance on Different Devices

We tested the effectiveness of our model on different hardware setups to ensure its reliability. These include an RTX4090 (GPU), an RTX3060 (GPU), an Nvidia Jetson NX 8G (edge device), and an Intel 10700k (CPU). To improve the performance in the CPU environment, we converted the model weights into the open neural network exchange (ONNX) inference framework. The results, as shown in Table IV, indicate that ELLK-Net outperforms general object detection models in terms of FPS across various hardware environments. Although the FPS of ELLK-Net is lower than those of the YOLOv3 and YOLOX models, its detection accuracy is significantly greater. Furthermore, after structure reparameterization, the inference speed on Nvidia Jetson NX is improved by approximately 48.7% FPS, which indicates that ELLK-Net is well suited for edge device deployment because it strikes a balance between detection accuracy and speed.

IV. DISCUSSION

In this section, we present additional experimental evaluations of ELLK-Net across diverse scenes and provide a

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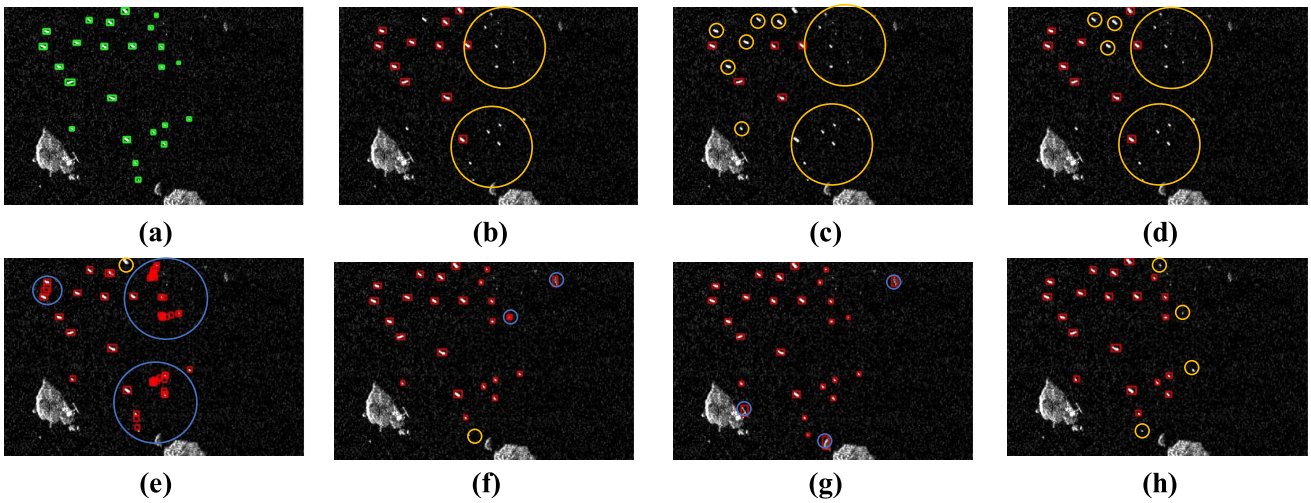


Fig. 13. Small-ship scene detection results of different methods on the SSDD dataset. Yellow circles represent missed detections, and blue circles represent false detections. (a) Ground truth. (b) Faster R-CNN. (c) FoveaBox. (d) Libra R-CNN. (e) RetinaNet. (f) YOLOv3. (g) YOLOX. (h) Our method.

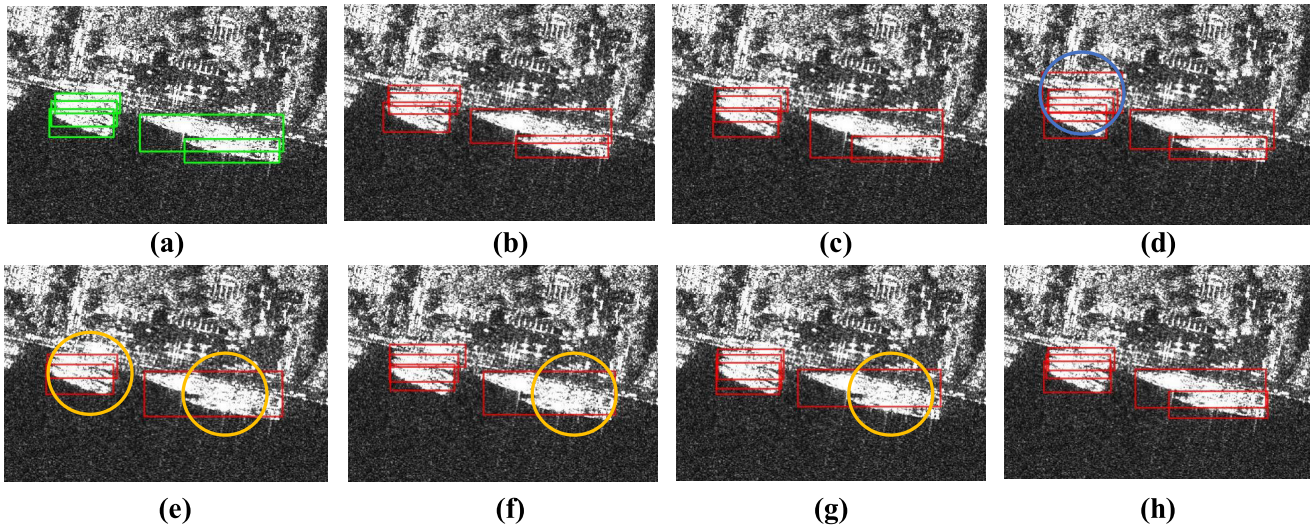


Fig. 14. Inshore interference scene detection results of different methods on the SSDD dataset. (a) Ground truth. (b) Faster R-CNN. (c) Fovea Box. (d) Libra R-CNN. (e) RetinaNet. (f) YOLOv3. (g) YOLOX. (h) Our method.

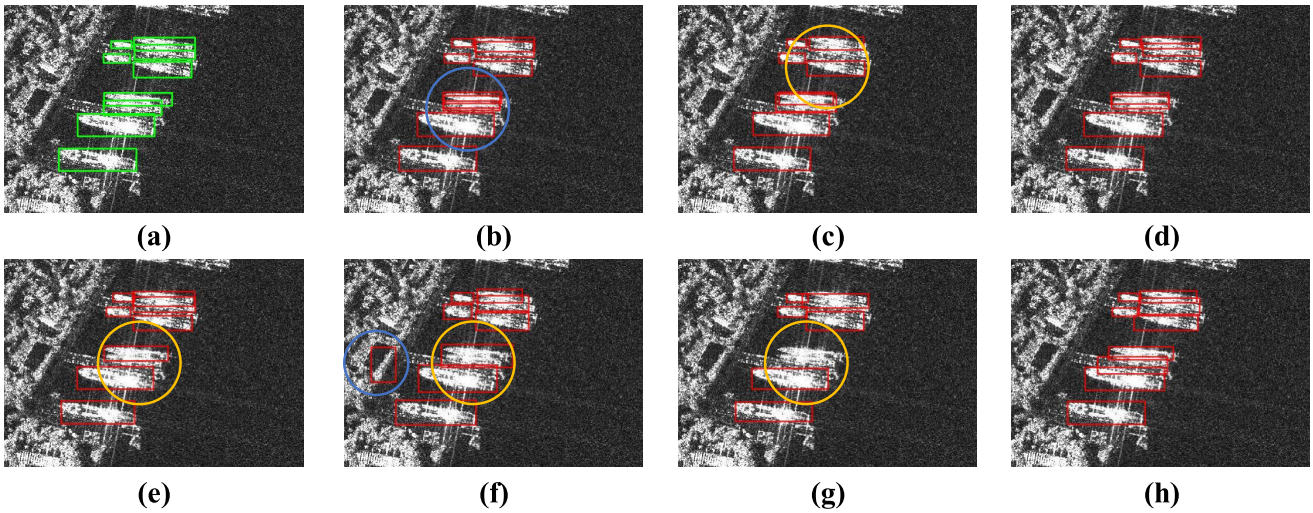


Fig. 15. Densely arranged ship scene detection results of different methods on the SSDD dataset. (a) Ground truth. (b) Faster R-CNN. (c) Fovea Box. (d) Libra R-CNN. (e) RetinaNet. (f) YOLOv3. (g) YOLOX. (h) Our method.

comprehensive analysis of the obtained results. To evaluate the generalizability of the proposed ELLK-Net model, we conduct validation experiments with rotation detection scenes and large-scene SAR images.

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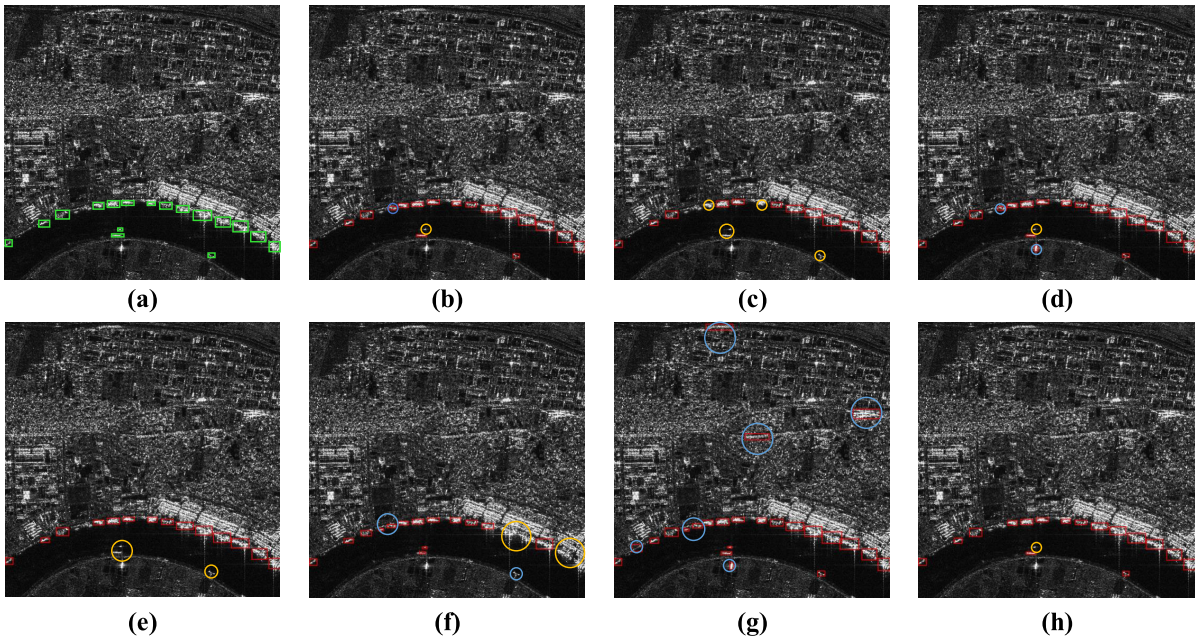


Fig. 16. Inshore interference scene detection results of different methods on the SSDD dataset. (a) Ground truth. (b) Faster R-CNN. (c) Fovea Box. (d) Libra R-CNN. (e) RetinaNet. (f) YOLOv3. (g) YOLOX. (h) Our method.

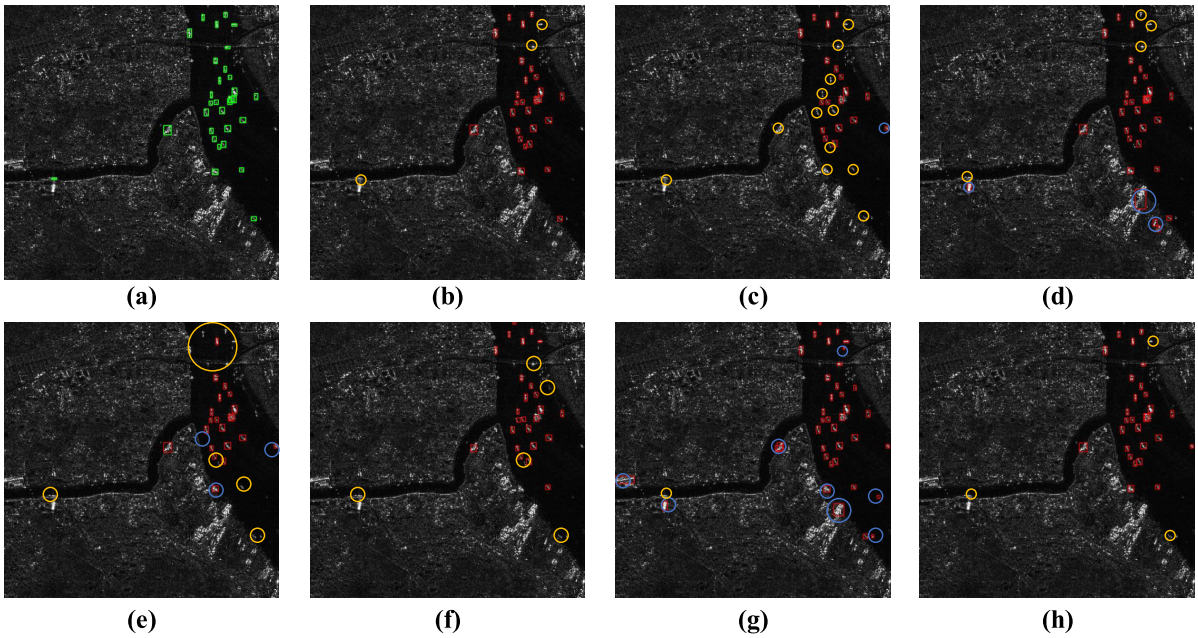


Fig. 17. Inland complex interference scene detection results of different methods on the HRSID dataset. (a) Ground truth. (b) Faster R-CNN. (c) Fovea Box. (d) Libra R-CNN. (e) RetinaNet. (f) YOLOv3. (g) YOLOX. (h) Our method.

A. Different Scenes of Evaluation

Inshore and offshore scenes are two complex scenes in SAR ship detection. In both cases, we evaluate the detection performance of the proposed and baseline methods. In inshore scenes, ground noise causes substantial interference, ships are densely distributed, and scattered interference may occur around ships. As Table V shows, compared with the baseline, the proposed method improves the AP, AP_{50}, and AP_{75} by 3.8%, 5.5%, and 3.2%, respectively, in inshore scenes. In offshore scenes with many scattered small ships, the proposed method increases the AP, AP_{50}, and AP_{75} by 3.4%, 0.4%, and 3.1%, respectively. The results show that ALKS-Net captures global information, models the long-range dependence between the background and objects, and enhances fine-grained multiscale feature expression. LKA-FEP enhances interlayer multiscale feature fusion and semantic information alignment through feature fusion between different layers, reducing the impact of the cross-overlap of ship targets and scattering noise on accuracy. ELLK-Net maintains detection accuracy for different-scale targets across scenes.

B. Evaluation of Rotation Detection Scenes

To test the generalization ability of our proposed ELLK-Net, we conducted experiments on the SSDD and HRSID

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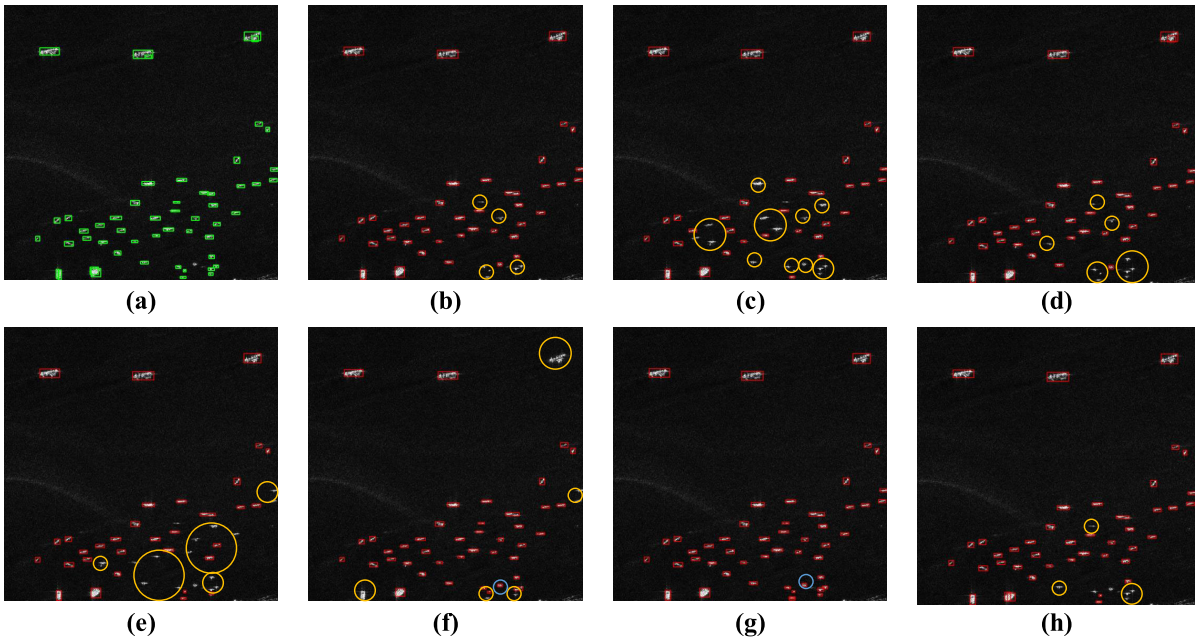


Fig. 18. Small-ship scene detection results of different methods on the HRSID dataset. (a) Ground truth. (b) Faster R-CNN. (c) Fovea Box. (d) Libra R-CNN. (e) RetinaNet. (f) YOLOv3. (g) YOLOX. (h) Our method.

TABLE IV

FPS INFERENCE BY DIFFERENT MODELS IN DIFFERENT HARDWARE ENVIRONMENTS (SSDD)

method	RTX 4090	RTX3060	Intel 10700k	Nvidia Jetson NX(8G)
FoveaBox [31]	85.6	28.6	7.1	3.2
Faster R-CNN [8]	98.1	26.5	5.6	2.6
Libra R-CNN [25]	98.9	26.2	5.9	2.5
RetinaNet [9]	89.8	27.6	6.9	3.2
YOLOv3 [30]	**324**	**62.3**	**43.1**	**21.2**
YOLOX [13]	177	45.9	21.6	9.4
Ours	99.3	25.6	8.6	3.9
Ours(REP)	110.6	29.1	10.9	5.8

TABLE V

COMPARISON OF DETECTION ACCURACY IN OFFSHORE AND INSHORE SCENES

scene	method	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
offshore	baseline	0.651	0.974	0.797	0.594	0.741	0.756
	ours	**0.685**	**0.978**	**0.828**	**0.641**	**0.761**	**0.799**
inshore	baseline	0.488	0.817	0.534	0.373	**0.650**	0.690
	ours	**0.526**	**0.872**	**0.566**	**0.439**	0.643	**0.725**

rotation annotation datasets by implementing rotating SAR target detection by adding an angle branch to the detection head. Our model was compared with other rotating object detectors, such as Oriented RCNN [34], Rotated Faster R-CNN, Rotated FCOS, Rotated RetinaNet, and S2anet [35]. The rotation detection results for the SSDD and HRSID datasets are shown in Fig. 19 and Table VI: on the SSDD dataset, ELLK-Net achieves a detection accuracy close to that of general rotating target detectors while significantly reducing the number of parameters and computations. Compared with the Oriented RCNN, the number of parameters is reduced by 62.5%, and the precision is improved by 5.5%. Compared

TABLE VI

COMPARISON OF ROTATION DETECTION METHODS ON SSDD AND HRSID DATASETS

method	Precision	Recall	AP_{50}	Flops(G)	Params(M)	FPS(img/s)
Oriented R-CNN [34]	0.848	**0.932**	**0.901**	83.5	41.3	80.3
Rotated Faster R-CNN	**0.934**	0.900	0.897	83.4	41.1	85.1
Rotated FCOS	0.710	0.925	0.891	72.6	41.4	91.7
Rotated RetinaNet	0.804	0.912	0.886	73.8	36.1	76.8
S2anet [35]	0.745	0.915	0.883	69.7	38.6	69.4
ours	0.903	0.925	0.897	**23.7**	**15.5**	**109.1**
Oriented R-CNN [34]	0.810	0.806	0.790	232.8	41.3	59.2
Rotated Faster R-CNN	0.803	0.809	0.791	211.3	41.1	62.9
Rotated FCOS	0.742	0.826	0.783	212.3	41.4	75.1
Rotated RetinaNet	0.702	0.834	0.785	209.1	36.1	58.8
S2anet [35]	0.745	0.818	0.789	196.2	38.6	51.0
Ours	**0.896**	**0.877**	**0.797**	**67.2**	**15.5**	**82.3**

with other anchor-free rotated object detectors such as Rotated FCOS, we significantly improve the precision. The results with respect to the HRSID dataset are shown in Table VI. For all the accuracy metrics, ELLK-Net achieves the best performance. Compared with the Rotated Faster R-CNN, the number of parameters is reduced by 61.7%, the number of computations is reduced by 68.2%, the precision is improved by 9.3%, and the recall is improved by 6.8%. FPS increased by 30.8%. Notably, the detection head part of our model was not specifically designed for rotating object detection but incorporated an angle branch to enable such functionality. This observation highlights the strong capability of ELLK-Net in feature extraction and fusion and its ability to generate feature maps suitable for higher-level rotating object detectors. Consequently, our method demonstrates excellent generalization performance in this domain. Furthermore, ELLK-Net has faster inference speeds than other rotating object detectors do, making it highly advantageous for practical deployment.

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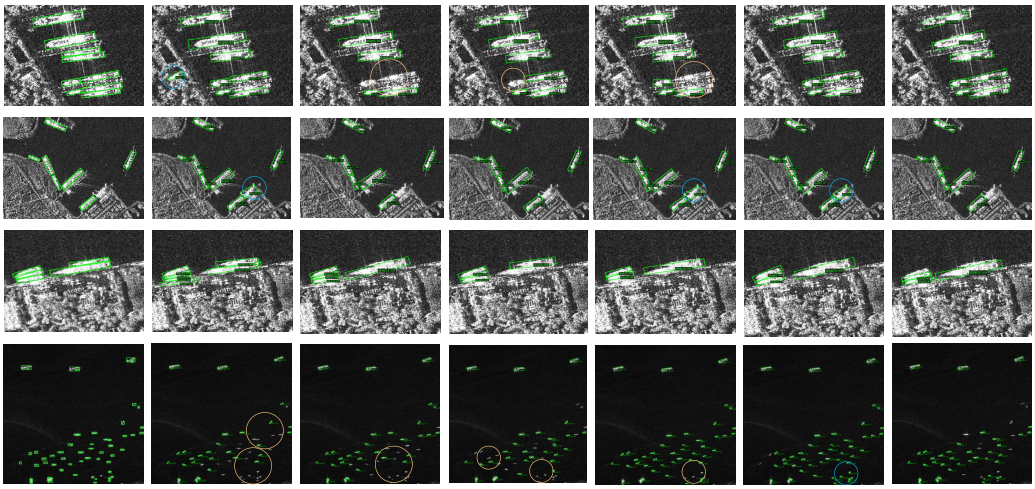


Fig. 19. Rotation detection results for the SSDD (the first to third rows) and HRSID (the fourth row) datasets. (a) Ground truth. (b) Oriented R-CNN. (c) Rotated Faster R-CNN. (d) Rotated FCOS. (e) Rotated RetinaNet. (f) S2anet. (g) Our method.

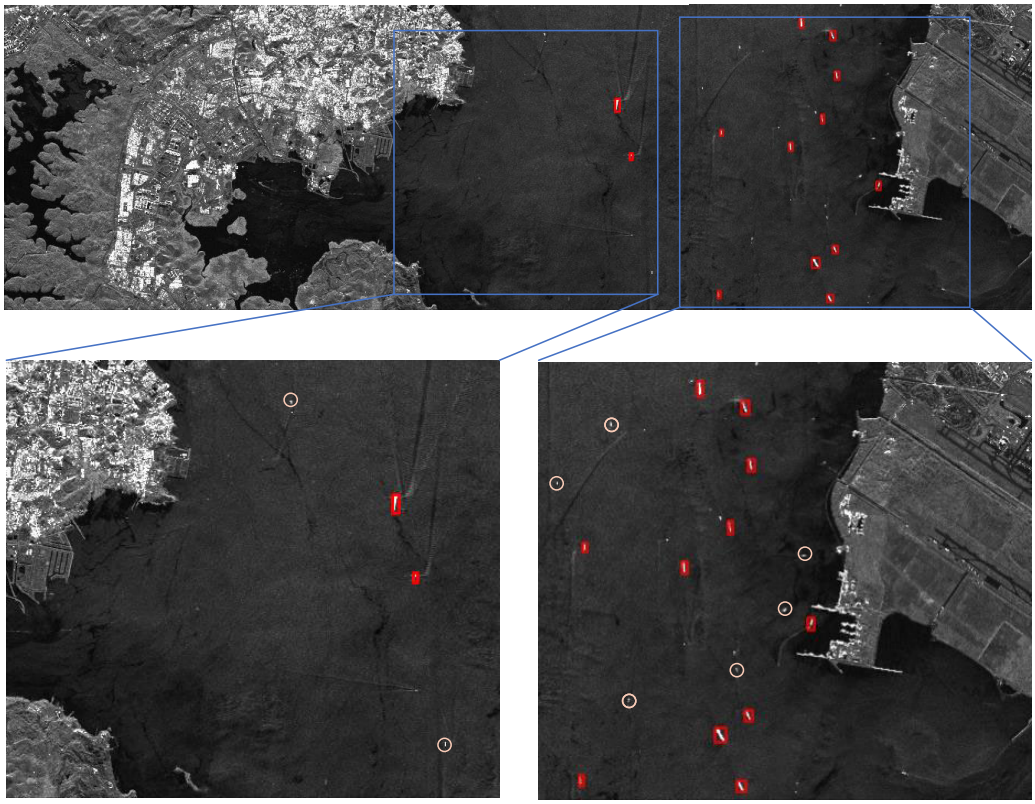


Fig. 20. Large-scene detection results for SAR images. The red box indicates the detection results, and the yellow circle indicates missed ships.

C. Verification on Large-Scene SAR Images

To test how well our method can be applied to large-scale scenes, we obtained a TerraSAR-X SAR image with dimensions of 1340 × 2500 from an open website. As shown in Fig. 20, our method was highly effective in processing SAR images of vast areas and accurately detected a significant number of prominent ship targets within these images. These results substantiate that our method exhibits robust generalization capabilities and consistently delivers satisfactory outcomes across diverse SAR scenarios.

V. Conclusion

We present ELLK-Net, an efficient and lightweight network for SAR ship detection via large kernel convolution for enhanced global information capture within the backbone network. It leverages the adaptive selection of large convolution kernels and an LKMA module to improve feature expression and interlayer fusion. To optimize the inference speed, the network employs structural reparameterization to decouple large kernel convolutions. ELLK-Net demonstrates high accuracy and speed on the SSDD and HRSID datasets

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for both horizontal and rotating SAR ship detection, making it suitable for edge device deployment. Future work will explore further optimization through quantitative and pruning techniques and hardware-specific joint optimization.

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